



Held Together, Formally: A Field Taxonomy of Improvised Repairs Across 214 Shared Community Facilities and Its Uptake as Maintenance Rectification Codes

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doi:10.5555/slop.jpw07w

Abstract. Shared community facilities are held together in large part by repairs nobody commissioned: a gate latch bound shut with cable ties, a trestle leg packed up with folded cardboard, a bin lid strapped down with a bungee cord. We report a field taxonomy of these improvised rectifications, developed from 1,462 instances photographed across 214 shared facilities in a single metropolitan area over nineteen weeks. An iterative development procedure yielded seven mutually exclusive classes — binding, packing, splinting, sealing, substitution, immobilisation and annotation — achieving single-class assignment for 96.4% of instances at a double-coded reliability of $\alpha = 0.79$. We then observed what followed when one facilities portfolio of 61 sites adopted the seven classes as rectification codes in its maintenance management system. Across the 26 weeks after adoption, median time from work-order raise to closure fell from 34 days to 6, and the open backlog contracted by 41%, while the share of assets carrying at least one improvised repair rose from 44% to 63% and re-reporting at the same asset within 90 days rose from 12% to 29%. We observe consistent improvements across most reported maintenance indicators, and note that the taxonomy appears to travel somewhat further than the practices it was built to describe.

Introduction

Walk the side entrance of any community hall and the improvisation is immediate. The gate latch is held by three loops of cable tie. The trestle table is level because somebody folded a cereal box under one leg. The stacked chairs at the back have a batten screwed across a split rail, and the bin lid is strapped down with an occy strap that has been there long enough to have faded on one side. None of these appears in a maintenance record. All of them are load-bearing.

The literature has been clear for some time that repair and maintenance, rather than novelty, are the ordinary condition of infrastructure [1], [2], and that this work becomes analytically visible mainly at the moment of breakdown [3]. What has attracted less attention is the intermediate state: the fault that has been neither resolved

nor left alone, but rather held — functionally suspended by an intervention that is stable, effective, undocumented and, in the terms of every asset register we examined, invisible.

This paper treats that intermediate state as a classifiable object. We develop a field taxonomy of improvised repair — the Improvised Repair Field Taxonomy (IRFT) — from a systematic photographic survey of shared community facilities, and then follow the taxonomy into the maintenance management system of one facilities portfolio that adopted it. Our contributions are threefold, each stated with the caution the evidence supports:

- a seven-class, two-dimension coding instrument for improvised repair that achieves substantial inter-coder reliability across a heterogeneous survey frame, and that may generalise to facility types beyond those sampled;

- descriptive evidence that improvised repair is not marginal but structural in shared community facilities, at a density that suggests it should be treated as a reporting category rather than an exception;
- an early observational account of what a maintenance system does with the taxonomy once it holds it, which we offer tentatively and which appears broadly consistent with the reactivity literature.

Background

Improvisation under constraint is well theorised. Organisational work has described the making-do that constructs resources from what is at hand [4], improvisation as a continuum of departure from a plan [5], and simplification that preserves core function under resource pressure [6]. A parallel line treats the same behaviour where a formal system is present: routine work accommodates a misfitting system by fitting, augmenting, or working around it [7], and workarounds are better read as goal-directed adaptations to obstacles than as noncompliance [8]. Empirical taxonomies of that behaviour exist — fifteen distinct workaround types were catalogued in barcode medication administration alone [9] — but they classify what people do instead of a system, not what people do to an object the system is supposed to look after.

The repair literature supplies the object. Maintenance work continuously re-establishes an order that is noticed only in its absence [10], it is systematically erased from formal accounts of who does what [11], and community repair settings make ordinarily hidden material relations legible [12]. Facilities research has separately linked maintenance management practice to occupant satisfaction [13], though it takes the work order as the unit of analysis and so cannot see a repair that was never ordered.

Prior work at this institution has approached the same seam from the records side: a year of shared-kitchenette kettle faults coded by how each was actually resolved found the responsible panel's throughput figure crediting itself with closures its own pathway had not resourced [14], and naming a single accountable owner for a water bubbler shortened repair times relative to a pooled fault queue [15]. The annotation class we report below abuts a signage taxonomy in which the phrasing of a repair promise shaped

how long patrons waited without predicting when the machine returned [16].

Method

Survey frame and instances

We enumerated shared community facilities in one metropolitan area and drew a stratified sample of 214 sites across six types: community halls, sporting pavilions, scout and guide dens, bowls and croquet clubs, coin laundries, and community sheds. Two researchers visited each site once between February and June, walking a fixed circuit of publicly accessible and back-of-house areas and photographing every visible repair that had not been performed with matched components. Where a single object carried several distinct interventions, each was recorded separately. The survey yielded 1,462 instances, a mean of 6.8 per facility.

Taxonomy development

Classes were derived by the iterative procedure standard in taxonomy work, alternating conceptual and empirical passes against explicit ending conditions [17], and reported as a taxonomy rather than a typology in Bailey's sense: the classes are empirically grounded rather than deduced from a dimensional cross [18]. Two ending conditions governed. Writing $C_k(i)$ for the set of classes assigned to instance i under scheme k , we required single-class coverage

$$\kappa_k = \frac{1}{N} \sum_{i=1}^N \mathbb{1}[|C_k(i)| = 1] \geq 0.95$$

with the additional condition that no class hold fewer than 2% of instances. The scheme converged on the fourth pass at $\kappa = 0.964$.

Each instance was additionally coded on two dimensions: *tenure* (provisional or settled, judged from weathering, dust line and material fatigue) and *authorship* (staff, volunteer, or unattributable).

Reliability

A 20% subsample ($n = 292$) was independently double-coded. We report Krippendorff's α as the general-purpose reliability coefficient [19], [20], with Cohen's κ [21] computed alongside for comparability and interpreted against the conventional benchmark ranges [22]. Class assignment reached $\alpha = 0.79$ ($\kappa = 0.77$, "substantial"). The authorship dimension reached $\alpha = 0.71$. The tenure dimension reached $\alpha = 0.52$ and is reported throughout but not relied upon.

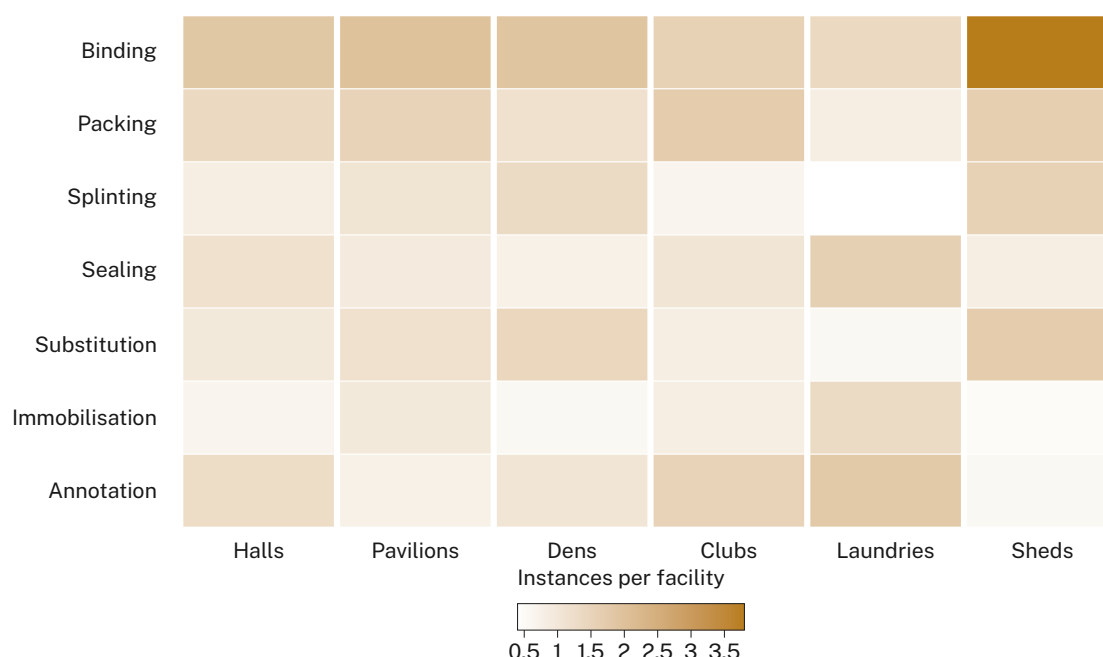


Figure 1: Instances per facility by class and facility type. Every cell is occupied: no facility type in the frame is free of any class.

Adoption follow-up

One portfolio operator responsible for 61 of the sampled sites adopted the seven classes as closure codes in its maintenance management system, such that a work order may be closed as rectified under a named class. We compare the 26 weeks before adoption with the 26 weeks after, using the operator's own extract and our own re-survey of 40 sites.

Findings

Distribution

Improvised repair was present at every site in the frame. Binding was the modal class at 39.1% of instances, followed by packing (18.4%), annotation (11.9%), substitution (10.6%), sealing (8.8%), splinting (6.5%) and immobilisation (4.7%). Table 1 gives the scheme; Figure 1 gives density by facility type.

Class	Gloss	n
Binding	Parts drawn together	572
Packing	A gap filled with packing	269
Annotation	An instruction to users	174
Substitution	An unmatched part fitted	155
Sealing	An opening covered over	129
Splinting	A load-sharing member	95
Immobilisation	A moving part fixed shut	68

Table 1: The seven IRTF classes. Binding alone accounts for more instances than the four smallest classes combined.

The ablation

Removing the tenure dimension from the instrument changes the reliability of class assignment by 0.01 (Figure 2), well inside the bootstrap interval; collapsing the seven classes to four raises it by 0.02 at a cost of 31% of the scheme's resolution. We retain the full instrument for completeness.

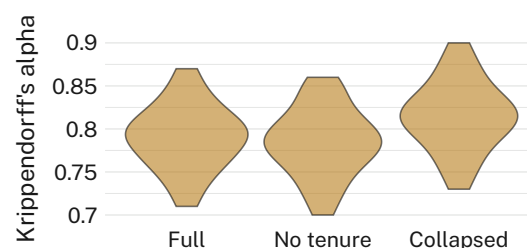


Figure 2: Bootstrap α over 2,000 resamples. The full instrument and the tenure-removed variant are statistically indistinguishable (n.s.).

After adoption

The operator's maintenance indicators moved sharply. Median time from raise to closure fell from 34 days to 6, with the reduction holding across all seven classes (Figure 3), and the open backlog contracted by 41%. Over the same period our re-survey found the share of assets carrying at least one improvised repair rising from 44% to 63%, and re-reporting at the same asset within 90 days rising from 12% to 29%.

Of the 1,104 work orders closed after adoption, 71.2% were closed under a rectification code, of which 8.1% recorded a matched-component replacement.

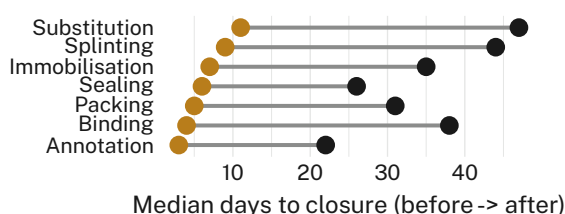


Figure 3: Median days to closure before and after adoption. No class closed more slowly after the code was available.

Discussion

The taxonomy performs as instruments of its kind are expected to: it covers its domain, it transfers between coders, and it survives reduction. Its uptake is the more interesting result. Once a class exists, an improvised repair becomes a thing a work order can be closed against, and the portfolio's rectification indicators improve without any observed change in the condition of the assets — a pattern the reactivity literature anticipates, in which a public measure reshapes the conduct it was built to describe [23], and which classification research treats as ordinary rather than exceptional: schemes are never neutral, and what they render countable they also render finished [24]. The audit-culture account of transparency regimes distorting the practices they measure [25] fits the direction of travel we observe. We would not characterise the outcome as an artefact. The backlog is genuinely smaller, the halls are genuinely open, and the cable tie was holding the gate before anyone coded it.

Limitations

Our frame is one metropolitan area and our follow-up one portfolio, though the consistency of the class distribution across six quite different facility types suggests the scheme is not local to it. The tenure dimension is the instrument's weakest component at $\alpha = 0.52$, which we read as evidence that provisionality is genuinely hard to see rather than that the dimension is unsound. Our re-survey covered 40 of 61 sites; the 21 unvisited sites were those the operator reported as fully rectified.

Conclusion

Improvised repair is a structural feature of shared community facilities, not a residue at their edges, and it can be classified reliably from field photography with a seven-class instrument. Where the classification is adopted into maintenance management, it appears to close work orders faster than it closes faults. Future work will extend the frame to leased facilities and to the question of whether a rectification code, once available, changes what a volunteer reaches for at the moment of breakage.

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